

	Research Interests
	I have academic and research experience in solving urban and regional planning problems through the application of geospatial and statistical tools, technologies, and policy-driven interventions. I am highly motivated to work on geospatial problems using machine learning, AI, and technology-driven methods, and I am eager to explore new technologies in this field.
	Education
2026 – Expected Jan 2027	Masters in Urban & Regional Planning , Faculty of Civil Engineering, Khulna University of Engineering & Technology (KUET)
2020 – 2025	Bachelor in Urban & Regional Planning , Faculty of Civil Engineering, Khulna University of Engineering & Technology (KUET) CGPA– 3.97 on 4 (All semesters). Position in Class– 1 st in all academic years (2020–2025).
2017 – 2019	Higher Secondary Certificate (Science) , RAJUK Uttara Model College, GPA – 5.00 on 5.00
	Thesis
	Bachelor's Thesis
Title	<i>Urban Growth Dynamics and Socioeconomic Drivers in Khulna: A Spatiotemporal Perspective</i>
Supervisor	Professor Dr. Md. Manjur Morshed
Description	An urban expansion analysis using Remote Sensing and Socioeconomic Indicators (2000–2024) integrating binary classification, spatial statistics geographic distribution tools including JavaScript and geographic information systems.
	Research & Publications (published as “Mst. Tasnia Nowrin”)
1	Al Kafy, A., Nowrin, M. T. , Altuwaijri, H. A., & Mukarram, M. M. T. (2025). <i>Advanced hybrid ensemble learning models for atmospheric humidity prediction: An environmental monitoring solution in arid regions</i> . <i>International Journal of Environmental Science and Technology</i> . DOI: https://doi.org/10.1007/s13762-025-06926-z
2	Nowrin, M. T. , Rahman, M., Hasan, M. M., & Hasan, M. M. (2026). <i>Price Hikes, Sustainability, and SDG Achievement Among Lower-Income Groups in Urban Bangladesh</i> . Manuscript previously submitted; under revision for resubmission.
3	Nowrin, M. T. & Mohiuddin, T. (2025, February). <i>Sustainable Desalination Technologies: A Solution to Coastal Water Scarcity in Bangladesh</i> (Poster presentation). Plannation 3.0, Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh.
4	Fattah, M. A., Rahman, M., Hasan, M. M., Hasan, M. M., Nowrin, M. T. , Morshed, S. R., Chakraborty, T., Al Kafy, A., Alsulamy, S., Khedher, K. M., & Shohan, A. A. A. (2023). <i>Sustainability amidst economic strain: A quantitative exploration of COVID-19 induced food insecurity, obesity, and inflation impacts on health and sanitation in urban slums</i> . (Unpublished manuscript)
	Professional Experience
Oct 2024	Intern , Tiller , Dhaka, Bangladesh
	- Contributed to the “Preparation of Risk Sensitive Database for Core Area of Rangpur and Sylhet District Town (SRDP)” project. - Assisted in preparing land-use zoning and Mouza maps scaling using GIS for risk-sensitive planning. - Gained hands-on experience with urban spatial data processing and risk-sensitive zoning practices.
2023 –	Research Assistant , Center for Environmental Research (CER)

	- Contributed to multidisciplinary research on environmental change, air pollution, transportation, and public health using GIS and statistical analysis (GEE, ArcGIS, Python). - Assisted in drafting manuscripts, data collection and processing, and climate and transport-related modeling to provide policy-relevant, evidence-based solutions for a sustainable future.
	Projects
○	Transportation Surveys <i>Traffic surveys, AutoCAD, GIS and Excel for intersection and mobility analysis.</i>
○	Regional Economics Analysis of Rajshahi Division <i>Used PCA, SPSS, and GIS to map economic disparities.</i>
○	Comparative Analysis of Smart Village Potential: Evaluating the Smartness of Botiaghata and Betaga Union <i>Field and questionnaire surveys, GIS and SPSS-based of digital infrastructure in rural Bangladesh.</i>
○	Assessment of Child Friendliness in Ispahani Colony, Khulna <i>Conducted statistical analysis, field and questionnaire surveys.</i>
○	Water Scarcity in Rajshahi Division: A feasibility Study for Sustainable Solutions <i>Meta-analysis and GIS-based rainwater harvesting and recharge strategy.</i>
○	Heritage Restoration – The Mosque City of Bagerhat <i>Developed DPP & ToR with feasibility, budgeting, impact analysis & stakeholder mapping.</i>
○	Sustainable Desalination Technologies: A Solution to Coastal Water Scarcity in Bangladesh <i>PV-RO technology planning with GIS mapping for coastal water security.</i>
	Skills
Coding	Python, JavaScript, C, R
Tools & Softwares	ArcGIS, Google Earth Engine (GEE), Google Earth Pro, SPSS, Excel, AutoCAD, SketchUp, MS Office
Techniques	Network Analysis, Remote Sensing, GIS-Based Modeling, Spatial Analysis, Machine Learning
Professional	Project Management, Stakeholder Engagement, Teamwork, Leadership, Communication under Pressure
Language & Proficiency	Bangla (native), English (proficient); undergraduate program conducted in English (KUET, WHED-listed). IELTS planned.
	Awards
2019–20	Dean’s Award for Academic Excellence, Faculty of Civil Engineering, Khulna University of Engineering & Technology (KUET) Technical and Compensatory Scholarship, Khulna University of Engineering & Technology (KUET)
2020–21	
2021–22	
	Leadership & Extracurricular
2025	Volunteer , Plannation 3.0, Khulna University of Engineering & Technology (KUET) Logistical Support, Event planning and team coordination.
2020 – 2024	Team Leader , Academic Lab Projects Led multiple projects in GIS and urban planning as well as contributed to surveying, research documentation, including literature reviews, technical reports, and academic presentations.
2020 – 2023	Co-Organizational Secretary , KUET Theatre Event planning and team coordination.